

CivicTwin: A Counterfactual Graph Learning Framework for Ex-Ante Urban Policy Comparison

Abhay Mahantesh Zalaki, Ayush Raj, and Prabin S M

Abstract—Ex-ante policy comparison asks which of several interventions a city should prefer, before any is enacted. We treat this as causal inference under spatial interference rather than as forecasting: an intervention in one neighbourhood moves rents next door, so the Stable Unit Treatment Value Assumption fails and an estimator which ignores this charges the neighbour response to the control condition. We propose the Causal Graph Counterfactual Exposure Network (CG-CEN), which splits the one-step-ahead prediction into a direct term evaluated at the unit and an interference term aggregated over its neighbours. The split is enforced by the architecture rather than by a penalty, since the direct term reads a temporal encoding no message-passing operator has touched. Contrasting each channel against its untreated counterfactual gives a treatment effect which vanishes at untreated units and an interference effect which vanishes on an edgeless graph. Over ten seeds and four scenarios on a synthetic city, CG-CEN reaches pooled MAE 8.88, against 11.87 for a spatial lag model and 20.03 for a graph-blind temporal MLP. Its margin over the spatial lag model is significant under intervention ($p = 0.0015$ to 0.0033) and not in the untreated equilibrium ($p = 0.127$). Deleting the edges costs 2.89 MAE and permuting them 4.88, so a wrong graph does more damage than none. We also identify a regime, in which spillover arrives once and stays, where no model gains from the graph at all.

Index Terms—Causal inference, spatial interference, SUTVA, graph neural networks, urban policy evaluation, counterfactual analysis, displacement pressure.

I. INTRODUCTION

URBAN policy is written for parcels and districts, and its consequences ignore those boundaries. Upzoning a corridor, relaxing parking minima or capturing land value to pay for transit changes rents and land values inside the treated zone, and it also changes them next door, through housing search, through commuting, and through what developers expect to happen next. For the question a planner actually asks, which is who is displaced and where, the diffusion is not a nuisance. It is the object of interest.

Interference and SUTVA. The potential outcomes framework rests on the Stable Unit Treatment Value Assumption [1], under which the outcome of a unit depends on the treatment given to that unit and on nothing else. Spatial spillover breaks the assumption outright. A tract next to an upzoned corridor

has been partially treated although it received no intervention, and an estimator which files it under the control condition will read the rent movement there as noise, or as evidence that the policy did less than it did. The literature on partial interference [2], [3], [4] handles this by letting the outcome depend on a low dimensional summary of the neighbours' assignments. We take that route and put the summary in a learned spatio-temporal model.

Comparison, not prediction. There is a second difficulty, less often discussed. Work on urban and mobility data usually poses the task as forecasting: what will rents be next month, what will traffic be next hour. Spatio-temporal graph networks are good at this [8], [9]. But an agency choosing between an inclusionary mandate and a land-value capture scheme is not asking what rents will be in 2031. It is asking which of several worlds, none of which has happened, is preferable, and for whom. A point forecast can be accurate and useless for that purpose, because it cannot be conditioned on an intervention which has not occurred.

We therefore fix the scope of the paper as ex-ante comparison. The quantity of interest is not $\hat{Y}_{i,t+1}$ on its own but the difference between $\hat{Y}_{i,t+1}$ under an intervention and under the untreated counterfactual, taken together with the same difference at the neighbours of i . Forecast accuracy enters as a necessary condition and not as the goal. A model whose conditional expectations are poorly calibrated cannot be trusted to produce differences between them.

CG-CEN: an introduction. Let $w_i \in \{0, 1\}$ record whether neighbourhood i receives the intervention. Write $\mathcal{N}(i)$ for the neighbours of i and W_{ij} for the row-normalised adjacency weights. The model predicts

$$\hat{Y}_{i,t+1} = \Phi(X_{i,t}, P_{i,t}) + \sum_{j \in \mathcal{N}(i)} W_{ij} \cdot \Psi(X_{j,t}, P_{j,t}), \quad (1)$$

where Φ carries the individual treatment effect and Ψ , once aggregated, carries the spatial interference effect. Nothing in (1) is unusual as an equation. What matters is where the two functions get their arguments from. The encoding fed to Φ is produced by a recurrent pass over the history of unit i alone, with no message passing anywhere in it, so there is no route by which the state of a neighbour can enter the direct term. The encoding fed to Ψ is evaluated at the neighbours and aggregated across edges, and it is the only such route. The separation is a property of the wiring, not of a penalty which might or might not bind.

Two identities follow at once, and we use both as regression tests. If $P_{i,t} = 0$ then the treatment effect at i is exactly zero, because the treated and untreated arguments of Φ coincide. If

The authors are with the Department of [DEPARTMENT], [INSTITUTION], [CITY, POSTCODE, COUNTRY] (e-mail: abhaymzalaki@gmail.com; [AUTHOR 2 E-MAIL]; [AUTHOR 3 E-MAIL]).

Manuscript submitted 6 September 2026. Corresponding author: A. M. Zalaki.

The synthetic generator, the benchmark harness and the scripts which produce every figure and table in this paper are available at <https://github.com/AbhayZ1/CivicTwin>.

the edge set is empty then the interference effect is exactly zero, because the sum in (1) is empty. Neither identity is learned and neither can drift.

Why a synthetic city. Credible spatial policy data with a known ground truth is scarce, and evaluating a spillover-aware estimator on observational data is close to circular: the spillover structure is exactly what is unknown, so a model which appears to exploit it cannot be told apart from one which has fitted confounding. We therefore build a generator whose process we specify completely, which lets us ask a question the observational setting cannot answer. When the spillover is known to be there, does the graph model find it? The answer turned out to depend on what kind of spillover it is, and that dependence is the main thing we have to report.

Summary of results. The contributions are as follows.

- Section II gives the architecture in which the direct and interference channels are separated by construction, together with the counterfactual contrasts which recover the two effects and the two degenerate cases in which they vanish exactly.
- Section II also revises the displacement pressure index. An index built only from growth rates cannot register a sustained policy effect: a permanent shift in level moves the growth rate once and then the index returns to its old path. Adding a cumulative burden term raises the measured contrast between scenarios by more than two orders of magnitude, from around 10^{-4} to around 1.7×10^{-2} .
- Section III reports the negative result. Under a generator in which the spillover is a single permanent shock, no model gained from the graph. Deleting the edges improved the network, permuting them changed nothing, and the learned spatial coefficient of the spatial lag model went to zero. We give the mechanism, which is that early permanent spillover is absorbed into the level of the receiving unit and is thereafter indistinguishable from a fixed effect.
- Section IV reports the benchmark on the repaired generator. CG-CEN is ahead of every baseline pooled over scenarios, and its margin over the spatial lag model is significant under intervention ($p = 0.0015$ to 0.0033) but not in the untreated equilibrium ($p = 0.127$). Permuting the edges is worse than deleting them.
- The generator, the four-model baseline suite, the ten-seed harness with confidence intervals and paired tests, and the topology ablations are released, along with a configuration profile which reproduces the degenerate regime so that the negative result stays auditable.

We put some weight on the third item. It is the kind of finding which normally disappears during model development, and had it disappeared here we would have published a paper claiming that a graph network captures urban spillover, on a benchmark which contained no spillover it could capture. Reporting where the boundary lies seems more useful than reporting only the side of it on which the method wins.

Related work. Three lines of work meet here. Causal inference under interference relaxes SUTVA by allowing the outcome of a unit to depend on the assignments of others.

Hudgens and Halloran [2] formalise partial interference within disjoint clusters, Ogburn and VanderWeele [3] give a graphical treatment, and Forastiere et al. [4] extend identification to observational networks through an explicit exposure mapping. We borrow the exposure mapping and place it inside a learned model rather than a parametric outcome regression.

Neural counterfactual estimation learns representations for treatment effect estimation, usually under SUTVA. Shalit et al. [11] give generalisation bounds for individual effects and Wager and Athey [12] give inference for forest-based heterogeneous effects. Closest to our setting is Ma and Tresp [13], who carry neural counterfactual estimation to networked interference. What we add relative to that line is the architectural separation of the two channels and the temporal dimension.

Spatio-temporal graph learning has been driven largely by traffic forecasting. Kipf and Welling [7] introduced the graph convolution we take as a reference point, and DCRNN [8] and STGCN [9] set the recurrent and convolutional templates. These are forecasting models fitted to observational series. Neither exposes a treatment channel or a counterfactual contrast, which is what comparison requires. On the measurement side we follow the combined housing and transportation framing of the H+T index [15], and our displacement indicators are informed by Zuk et al. [14], who document that public investment can raise accessibility and displacement risk at the same time. That tension is what the H+T index is built to expose.

Notation. Units are indexed by i and j and time steps by t . We write $X_{i,t} \in \mathbb{R}^F$ for the feature vector of unit i at step t , $P_{i,t} \in \{0,1\}$ for the policy assignment, d_i for the degree of i , and W_h for the length of the history window. Errors are always pooled over windows, units and features before they are reduced, so RMSE below is a root mean squared error and not a root mean square of per-window absolute errors. Standard deviations use the sample convention and confidence intervals are Student- t at the 95% level. All comparisons between models are paired on the seed.

II. THE MODEL

A. Graph and exposure

Let $G = (V, E, W)$ be an undirected weighted graph on $|V| = N$ units, a census tract in the empirical setting and a lattice cell in the synthetic one, with edges recording spatial adjacency. Write $\mathcal{N}(i)$ for the neighbours of i . The weight matrix is the row-normalised adjacency,

$$W_{ij} = \begin{cases} a_{ij} / \sum_k a_{ik}, & (i, j) \in E, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

so that $\sum_j W_{ij} = 1$ at every unit of positive degree. Normalising the rows turns the interference term into an average rather than a sum, which keeps its scale free of the degree and stops high degree units from dominating the objective.

Under SUTVA the potential outcome of unit i would be written $Y_i(P_i)$. Spatial spillover makes that inadequate, and we adopt instead the partial interference assumption, under which the outcome depends on the assignment of i and on the

neighbours' assignments only through a summary. We use the degree-normalised exposure

$$e_i = \sum_{j \in \mathcal{N}(i)} W_{ij} P_{j,t} = \frac{1}{d_i} \sum_{j \in \mathcal{N}(i)} P_{j,t}, \quad (3)$$

so $e_i \in [0, 1]$ is the treated fraction of the neighbourhood of i . The potential outcome becomes $Y_i(P_i, e_i)$, and the estimands separate into a direct effect at fixed exposure and an interference effect at fixed own treatment.

B. The two channels

Let $h_i \in \mathbb{R}^H$ be a temporal encoding of the history of unit i , built by a shared encoder followed by a gated recurrent unit [10],

$$z_{i,k} = \text{ReLU}(\mathbf{W}_{\text{enc}} \tilde{X}_{i,k} + b_{\text{enc}}), \quad k = t - W_h + 1, \dots, t, \quad (4)$$

$$h_i = \text{GRU}(z_{i,t-W_h+1}, \dots, z_{i,t}), \quad (5)$$

where $\tilde{X}$ denotes the standardised features of Section II-D. Note that (4) and (5) contain no message passing of any kind, so h_i is a function of the history of i and of nothing else.

Writing $s_i = [h_i \parallel P_{i,t}]$ for the state concatenated with the assignment, the prediction is

$$\hat{Y}_{i,t+1} = \underbrace{X_{i,t} + \text{MLP}_{\Phi}(s_i)}_{\Phi(X_{i,t}, P_{i,t})} + \underbrace{\sum_{j \in \mathcal{N}(i)} W_{ij} \text{MLP}_{\Psi}(s_j)}_{\text{interference}}. \quad (6)$$

The leading $X_{i,t}$ inside Φ is a residual anchor: the network is asked for the change from the last observation rather than for the level. This is not cosmetic. Without it the recurrent unit has to rebuild an absolute level through a saturating tanh bottleneck, and the near-linear part of the signal, which is simply that the next value resembles the last one, is destroyed on the way. Ablating the anchor and holding everything else fixed moves pooled MAE from 8.92 to 37.28, a factor of 4.2. Since $X_{i,t}$ is a function of the state of i , the anchor belongs to the direct term and does not contaminate the other one.

C. Recovering the two effects

Effects are recovered by contrasting each channel against its own untreated counterfactual. Put $s_i^0 = [h_i \parallel 0]$ and set

$$\tau_i = \text{MLP}_{\Phi}(s_i) - \text{MLP}_{\Phi}(s_i^0), \quad (7)$$

$$\eta_i = \sum_{j \in \mathcal{N}(i)} W_{ij} (\text{MLP}_{\Psi}(s_j) - \text{MLP}_{\Psi}(s_j^0)), \quad (8)$$

where τ_i is the estimated individual treatment effect and η_i the estimated spatial interference effect. If $P_{i,t} = 0$ then $s_i = s_i^0$ and $\tau_i = 0$ exactly. If $E = \emptyset$ then the sum in (8) is empty and $\eta_i = 0$ exactly. The additive identity $\hat{Y}_{i,t+1} = \Phi_i + \sum_j W_{ij} \Psi_j$ holds to floating point tolerance by construction. Algorithm 1 states the forward pass together with the contrasts.

Algorithm 1 CG-CEN forward pass with direct and spillover decomposition

Require: History $X_{i,t-W_h+1:t}$ for all $i \in V$; assignment $P_{i,t} \in \{0, 1\}$; neighbourhoods $\mathcal{N}(\cdot)$ with row-normalised weights W_{ij}

Ensure: Prediction $\hat{Y}_{i,t+1}$, effects τ_i and η_i

- 1: $\mu, \sigma \leftarrow \text{FITSCALER}(X_{\text{train}})$ {training split only}
- 2: $\tilde{X}_{i,\cdot} \leftarrow (X_{i,\cdot} - \mu)/\sigma$ for all i
- 3: **for** $k = t - W_h + 1$ **to** t **do**
- 4: $z_{i,k} \leftarrow \text{ReLU}(\mathbf{W}_{\text{enc}} \tilde{X}_{i,k} + b_{\text{enc}})$
- 5: **end for**
- 6: $h_i \leftarrow \text{GRU}(z_{i,t-W_h+1}, \dots, z_{i,t})$ {no message passing}
- 7: $s_i \leftarrow [h_i \parallel P_{i,t}]$, $s_i^0 \leftarrow [h_i \parallel 0]$
- 8: $\Phi_i \leftarrow X_{i,t} + \text{MLP}_{\Phi}(s_i)$ {residual anchor}
- 9: $\Omega_i \leftarrow \sum_{j \in \mathcal{N}(i)} W_{ij} \text{MLP}_{\Psi}(s_j)$
- 10: $\hat{Y}_{i,t+1} \leftarrow \Phi_i + \Omega_i$
- 11: $\tau_i \leftarrow \text{MLP}_{\Phi}(s_i) - \text{MLP}_{\Phi}(s_i^0)$
- 12: $\eta_i \leftarrow \sum_{j \in \mathcal{N}(i)} W_{ij} (\text{MLP}_{\Psi}(s_j) - \text{MLP}_{\Psi}(s_j^0))$
- 13: **return** $\hat{Y}_{i,t+1}, \tau_i, \eta_i$

D. Scaling

The panel spans several orders of magnitude. Median income is of order 10^4 and accessibility is of order 10^0 , so an unscaled squared error objective is governed by the large columns and the gradient carried by the rest is lost. We standardise each feature, $\tilde{X}_{\cdot,f} = (X_{\cdot,f} - \mu_f)/\sigma_f$, with μ_f and σ_f estimated on the training split only and reused unchanged on the test split, so that no distributional information crosses the temporal boundary backwards. Constant features are given $\sigma_f = 1$. Metrics are computed after inverting the transform, which keeps the errors in the original units and comparable across configurations.

E. Baselines

We compare against three baselines chosen to span the relevant hypothesis space. The *temporal MLP* flattens the history window and maps it through a two-layer perceptron. It cannot see the graph, and it isolates what the topology is worth. The *spatial lag model* is a learned form of the classical spatial autoregressive specification [5], [6],

$$\hat{Y}_{i,t+1} = X_{i,t} + \beta^{\top} \text{vec}(X_{i,t-W_h+1:t}) + \rho \odot \sum_{j \in \mathcal{N}(i)} W_{ij} X_{j,t}, \quad (9)$$

with a learned spatial coefficient ρ and a residual anchor matching (6), so that the comparison turns on functional form and not on parameterisation. The *persistence* model returns $\hat{Y}_{i,t+1} = X_{i,t}$, has no parameters and is never trained. It is the floor which any learned model must clear before a claim about spillover is worth making, and including it is what told us, at an earlier stage of this work, that both learned baselines were below it.

F. Two indices

Housing and transportation. Housing cost on its own understates what households in poorly connected neighbour-

hoods pay, because a low rent is often bought with a long commute. We define

$$H+T_{i,t} = \frac{\text{Rent}_{i,t} + \text{TransitCost}_{i,t}}{\text{MedianIncome}_{i,t}}, \quad (10)$$

with transit cost falling in accessibility,

$$\text{TransitCost}_{i,t} = \frac{c_0}{\max(\text{Accessibility}_{i,t}, \epsilon)^\kappa}, \quad (11)$$

for a base cost c_0 and an elasticity κ . This is the mechanism a rent-to-income ratio misses. A transit investment which raises accessibility lowers $H+T$ even where it raises rent, and whether the net movement is progressive is then a question the index can be asked.

Displacement pressure. The index blends adverse movements in rent, in low income share and in accessibility,

$$\begin{aligned} \text{DPI}_{i,t} = & \alpha \left(\frac{\Delta \text{Rent}_{i,t}}{\text{Rent}_{i,t-1}} \right) + \beta \left(1 - \frac{\text{LowIncShare}_{i,t}}{\text{LowIncShare}_{i,t-1}} \right) \\ & + \gamma (\Delta \text{Access}_{i,t}) + \delta B_{i,t}, \end{aligned} \quad (12)$$

in which LowIncShare abbreviates the low income share and Access the accessibility of the unit. The first three terms are the usual growth-rate form. The fourth is a correction, and the reason for it is worth stating plainly. A growth-rate index cannot see a sustained level difference. A policy which raises rents permanently by six per cent moves the growth rate in the single period in which the shift happens, after which the index returns to the path it was already on. In our runs this left every policy contrast at around 10^{-4} , present in the arithmetic and absent in any useful sense. We therefore add a cumulative burden term measured against the initial burden of the same unit,

$$B_{i,t} = \frac{H+T_{i,t}}{\max(H+T_{i,0}, \epsilon)} - 1, \quad (13)$$

which is zero at $t = 0$ and keeps a permanent offset under a permanent shift. Setting $\delta = 0$ recovers the growth-rate form exactly, and we pin that with a test so the original specification stays checkable.

Both indices are constructed comparative indicators over a controlled panel. Neither is a calibrated forecast of realised displacement, and we use them only to rank scenarios against a common baseline.

III. THE SYNTHETIC CITY

A. Graph and dynamics

The city is a 4×4 lattice of $N = 16$ units with four-connected adjacency, which gives 24 undirected edges, 48 directed entries and degrees between two and four. Each unit carries land value, rent, population, median income, housing units and accessibility, initialised with multiplicative Gaussian heterogeneity around city-wide bases of 100, 50, 2000, 60,000, 900 and 1.0 respectively. Generation is seeded and reproducible.

The dynamics are deliberately transparent. Accessibility shocks raise land values, land value growth feeds rent growth after a lag, rent pressure damps population, income and housing growth, and neighbour terms diffuse land and rent

across the edges. A designated set $S = \{0, 1, 4, 7\}$ receives a permanent accessibility uplift from $t \geq 3$, standing in for a transit investment.

B. A negative result: the static intercept

Our first generator used only that permanent shock, and under it no model benefited from the graph. We give the evidence before the explanation.

- The spatial lag coefficient of (9) converged to $\rho \approx 0$ across seeds, taking values -0.05 , -0.07 and 0.02 .
- CG-CEN with the edges deleted was *better* than CG-CEN on the real graph, 14.16 against 14.82 MAE.
- Permuting the edges was indistinguishable from the true topology, with p between 0.70 and 0.93.

The mechanism is instructive, and once seen it is obvious. A shock applied once at $t = 3$ to a fixed set of units means that by the evaluation window, which begins at $t = 12$, the neighbour contribution at each unit has settled to a constant. That constant is folded into the level of the unit and is recovered perfectly by the residual anchor $X_{i,t}$ of (6). Put differently, spillover which is both permanent and early is observationally the same thing as a unit fixed effect. There is nothing left for a graph to carry, because the information has already arrived by another road. We call this the static intercept, and we suspect it is a common and under-reported failure of spatio-temporal benchmarks which inject spillover as a single regime change.

C. Dynamic shocks

The repair is to keep the spillover moving. At each step every unit independently receives an accessibility pulse with probability $p_{\text{pulse}} = 0.22$, of magnitude drawn uniformly from $[0.5\sigma_p, 1.5\sigma_p]$ with $\sigma_p = 1.2$,

$$u_{i,t} \sim \text{Bernoulli}(p_{\text{pulse}}) \cdot \mathcal{U}(0.5\sigma_p, 1.5\sigma_p). \quad (14)$$

The pulse enters the observed accessibility of the unit at t , and, which is the point of the construction, it reaches the rent growth of the *neighbours* at $t + 1$,

$$g_{i,t}^{\text{rent}} += \lambda_p \sum_{j \in \mathcal{N}(i)} W_{ij} u_{j,t-1}, \quad (15)$$

with $\lambda_p = 0.22$. Because the pulse is transient and random, a pulse at a neighbour of i cannot be inferred from the history of i . It is new information which travels along an edge and nowhere else. Accessibility is observed, so the pulse at j is visible at time t to a model which can read across the edge, and it predicts the rent of i at $t + 1$. This is what makes the graph necessary rather than merely available.

D. Carrying capacities

The first generator applied purely multiplicative growth with nothing to equilibrate it, and this produced a second defect, worse than the first. Rents compounded without bound. As the rent-to-income ratio grew, the rent pressure term in the income update drove income sharply down, and income met its hard floor. In the unrepaid generator income reached the

floor at $t = 52$, and by $t = 100$ every one of the sixteen units was pinned there while mean rent had compounded to about 4×10^{10} . Any statistic taken past $t \approx 52$ was measuring the clamp and not the economy.

We add a logistic carrying capacity. Each growth rate is multiplied by a headroom factor which vanishes as the state approaches its capacity,

$$\text{headroom}(x, K) = \max\left(0, 1 - \frac{x}{K}\right), \quad (16)$$

with land capacity 12 times the base land value, income capacity 4 times the base income, and a rent capacity tied to contemporaneous ability to pay, $K_R = 0.012 \times \text{Income}_{i,t-1}$. Rent and income also carry mean reversion, towards $0.55K_R$ and towards the base income, at rates 0.05 and 0.02.

The economic reading is a supply response: as rents approach the local affordability ceiling, further growth is choked off by construction, by vacancy and by out-migration. The numerical effect is that the economy settles instead of diverging. Under the revised generator no unit reaches any hard floor through $t = 100$ on any of the ten seeds. Mean rent settles at about 1.62×10^3 , income at about 1.50×10^5 and H+T at about 0.011. Figure 1 shows the trajectories.

E. Scenarios

We define four scenarios, applied to the boundary S as an intervention on the generated panel. All four are simple and additive, so that the direction each one should move the indices is not in doubt.

- **Baseline.** No intervention. Supplies the untreated counterfactual against which the others are differenced.
- **Market-led development.** Upzoning with no affordability condition attached: land values up 4% and rents up 6% per unit intensity from $t \geq 2$. Should raise displacement pressure.
- **Inclusionary housing.** New development tied to affordable set asides: rents down 6% and stock up 8% per unit intensity from $t \geq 1$. Should lower pressure.
- **Land-value capture.** Taxation of uplift recycled into affordability: land values up a damped 2% and rents down 5% per unit intensity from $t \geq 2$. Should lower pressure, with a different land and rent signature from the inclusionary case.

Intensities of 0.8, 0.6 and 0.7 are configuration parameters and are not fitted. For each of ten seeds we export all four panels with the adjacency, the edge index, the low income share and the treated boundary, giving forty datasets.

IV. EXPERIMENTS

A. Protocol

Each run generates a seeded city of $N = 16$ units over $T = 18$ steps, applies one of the four scenarios, and forms rolling history windows of length $W_h = 4$ with next-step targets. This gives fourteen windows. The split is temporal, the earliest 70% for training and the rest for testing, with no shuffling, so nine windows train and five are held out and no future information

reaches the training set. The scaler of Section II-D is fitted on the training windows alone.

We model the three channels which carry the spillover, namely land value, rent and accessibility. Every model is constructed under an identical seed immediately before instantiation, so that a comparison between architectures, or between topology conditions, reflects what is being varied and not the draw of the initial weights. Learned models train for 250 epochs with Adam at learning rate 10^{-3} under a squared error objective. Persistence has no parameters and is not trained. Metrics are computed after inverting the scaler.

The design is ten seeds by four scenarios by four models for the main sweep, together with ten seeds by four scenarios by two ablated topologies for CG-CEN. This is 240 model instances of which 200 are trained, the remaining forty being persistence.

B. Accuracy

Table I gives MAE and RMSE by scenario and Figure 3 summarises the pooled comparison. Pooled over the four scenarios on the real graph, CG-CEN reaches $\text{MAE} = 8.88 \pm 2.40$, against 11.87 ± 3.12 for the spatial lag model, 20.03 ± 7.97 for the temporal MLP and 27.90 ± 1.27 for persistence. The RMSE ordering is the same, at 13.81, 19.30, 34.54 and 37.54, so the advantage is not an artefact of a few well-predicted units and survives a loss which weights large residuals more heavily.

The margin over the graph-blind MLP is large and significant in every scenario, with mean differences between -10.25 and -12.45 MAE and p between 9.4×10^{-4} and 5.8×10^{-3} . The margin over persistence is larger again, with $p < 2 \times 10^{-8}$ throughout. We treat the persistence comparison as a sanity floor and not as a result.

C. Where the graph pays

The comparison against the spatial lag model is the substantive one, since that model is itself graph-aware and differs from CG-CEN chiefly in functional form. Here the result is conditional, and Table II gives the condition rather than the pooled average.

CG-CEN is ahead by roughly 3.3 MAE under each intervention, with paired effect sizes between -1.25 and -1.42 , and its advantage of 1.94 MAE in the untreated equilibrium does not reach significance at ten seeds.

We think this boundary is the most useful thing in the paper. In a static equilibrium the spatial structure of the system is summarised well enough by a linear autoregressive lag, and the extra capacity of a nonlinear message-passing model buys nothing which survives the variance between seeds. Only when the interventions inject treatment-dependent perturbations which propagate does the interference channel have anything to work on. A paper which reported only the pooled comparison would have claimed a general forecasting advantage which the per-scenario decomposition does not support.

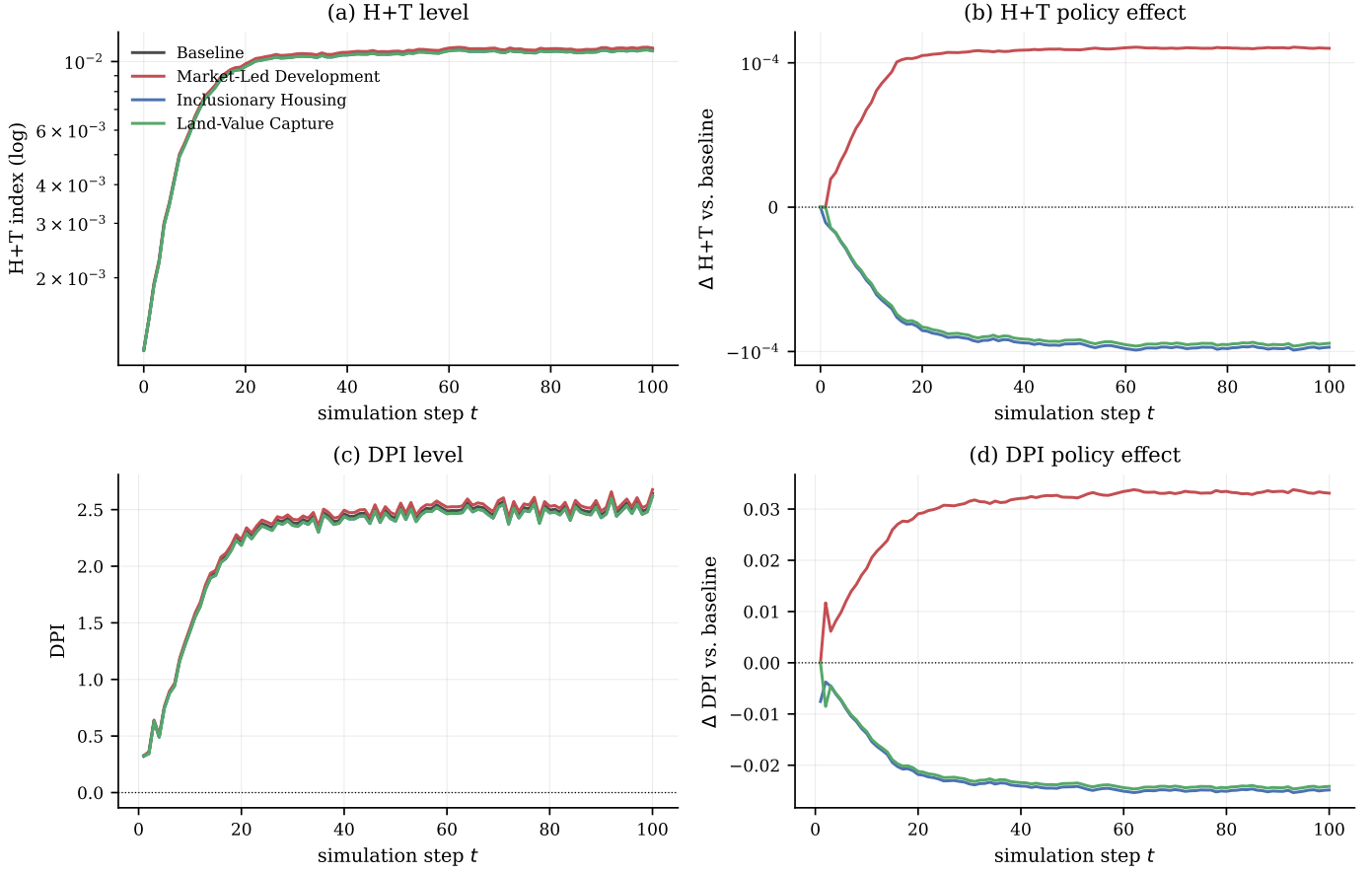


Fig. 1. Scenario trajectories over $t = 0 \dots 100$, city-mean, seed 1. Panels (a) and (c) give the levels of the H+T and DPI indices. Panels (b) and (d) give each intervention as a difference from the untreated baseline, which is the comparative quantity of interest; the inclusionary and land-value-capture curves lie close together and the latter is drawn over the former. Under the carrying capacities of (16) no unit reaches a hard state floor over the horizon, and the policy effects persist as level offsets instead of the single-step transients produced by a growth-rate index.

TABLE I
FORECAST ACCURACY BY POLICY SCENARIO ON THE REAL GRAPH

Scenario		CG-CEN (ours)	Spatial Lag	Temporal MLP	Persistence
Baseline	MAE	9.90 $\pm$ 2.23	11.84 $\pm$ 3.23	20.15 $\pm$ 8.57**	27.87 $\pm$ 1.30***
	RMSE	15.48 $\pm$ 2.99	19.27 $\pm$ 5.08	35.00 $\pm$ 15.90	37.49 $\pm$ 1.92
Market-led development	MAE	8.67 $\pm$ 2.95	12.01 $\pm$ 3.28**	21.12 $\pm$ 8.56**	28.27 $\pm$ 1.32***
	RMSE	13.76 $\pm$ 4.72	19.60 $\pm$ 5.17	36.78 $\pm$ 15.94	38.20 $\pm$ 1.98
Inclusionary housing	MAE	8.65 $\pm$ 2.37	11.80 $\pm$ 3.23**	19.45 $\pm$ 7.93**	27.70 $\pm$ 1.29***
	RMSE	13.32 $\pm$ 3.63	19.14 $\pm$ 5.08	33.24 $\pm$ 14.95	37.17 $\pm$ 1.88
Land-value capture	MAE	8.30 $\pm$ 1.99	11.83 $\pm$ 3.24**	19.40 $\pm$ 7.94***	27.77 $\pm$ 1.29***
	RMSE	12.69 $\pm$ 2.98	19.18 $\pm$ 5.10	33.14 $\pm$ 14.99	37.31 $\pm$ 1.88
Pooled over scenarios (MAE)		8.88 $\pm$ 2.40	11.87 $\pm$ 3.12	20.03 $\pm$ 7.97	27.90 $\pm$ 1.27

Values are mean $\pm$ standard deviation of the pooled test-set error over ten seeds; best per row in bold. Markers give a two-sided paired t -test of each baseline against CG-CEN on MAE (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$), pairing on the seed.

D. Topology ablation

To establish that the advantage comes from the graph and not from capacity, we refit CG-CEN under two counterfactual topologies with everything else held fixed, and Table III reports the outcome. Under *no edges* the interference term of (6) is identically zero. Under *shuffled* the edge index is remapped through a random permutation of the units, which preserves

the degree sequence and the edge count and destroys the correspondence between the graph and space.

Deleting the graph costs 2.89 MAE, a relative loss of 32.6%. Permuting it costs 4.88, a loss of 55.0%, significant in all four scenarios. The per-scenario breakdown is in Figure 4, and the one scenario in which edge deletion falls short of significance is the untreated baseline, at $p = 0.080$, which is consistent with the previous subsection.

TABLE II
CG-CEN AGAINST THE SPATIAL LAG MODEL, BY SCENARIO

Scenario	ΔMAE	t	p	d_z
Baseline	-1.94	-1.68	0.127	-0.53
Market-led	-3.34	-4.14	0.0025	-1.31
Inclusionary housing	-3.15	-3.96	0.0033	-1.25
Land-value capture	-3.53	-4.50	0.0015	-1.42

A negative ΔMAE favours CG-CEN. Ten seeds, paired on the seed; d_z is the paired effect size.

TABLE III
TOPOLOGY ABLATION FOR CG-CEN

Topology	MAE	Loss	p vs. real
Real graph	8.88	—	—
No edges	11.77	+2.89	0.005–0.008 (3 of 4)
Shuffled edges	13.76	+4.88	0.0001–0.018 (4 of 4)

Pooled over ten seeds and four scenarios. The final column gives the range of per-scenario p -values against the real graph and, in brackets, the number of scenarios reaching $p < 0.05$.

The observation which matters is that shuffled is worse than no edges. Were the model using message passing as a generic smoother, a random graph would serve about as well as the true one and both would sit between the real graph and the edgeless case. Instead a structurally plausible but factually wrong topology does more damage than supplying none. That is the signature of a model which has learned the particular adjacency: it propagates neighbour information with confidence, and when the information comes from the wrong neighbours the confidence is what hurts.

For contrast, the same ablation under the static-shock generator of Section III-B put no edges at 14.16 against 14.82 for the real graph, the edgeless model being ahead, with shuffled indistinguishable from real. The ablation is therefore not a confirmation of the final model. It is the diagnostic which separated a working benchmark from a degenerate one, and we would encourage treating the shuffled condition as routine rather than optional.

E. The two channels

Figure 2 shows the estimated effects on the lattice. The treatment effect concentrates on the treated boundary, as it must, since τ_i is identically zero at untreated units by (7). The interference effect spreads outward from the treated set and decays in graph distance, and it is exactly zero at units with no treated neighbour. Both are structural guarantees rather than learned behaviour, and we check them as unit tests along with the additive identity.

F. Scorecard

Table IV ranks the scenarios by displacement pressure mitigation. With the burden term of (12) active the contrasts are legible. Market-led development raises DPI by 0.0172 ± 0.0007 relative to baseline, while inclusionary housing and land-value capture lower it by 0.0131 ± 0.0006 and

TABLE IV
POLICY SCORECARD, RANKED BY DISPLACEMENT PRESSURE MITIGATION

Rank	Scenario	ΔDPI	$\Delta\text{H}+\text{T}$
1	Inclusionary housing	-0.0131	-4.7×10^{-5}
2	Land-value capture	-0.0125	-4.5×10^{-5}
3	Baseline	0	0
4	Market-led	+0.0172	$+6.1 \times 10^{-5}$

Negative values indicate that the intervention reduces the index relative to the untreated baseline. Means over ten seeds; the standard deviation of ΔDPI is below 8×10^{-4} in every case.

0.0125 ± 0.0005 . The two mitigating policies are close, and are ordered consistently across seeds with inclusionary housing slightly ahead, which is unsurprising since it acts on rent and on stock while land-value capture acts on rent alone.

Under the growth-rate form the same contrasts were of order 10^{-4} and returned to zero within one step of the intervention. We stress that this is a change in what the index measures and not in the simulated economy. The panels are identical and only the reading differs.

V. DISCUSSION AND OPEN QUESTIONS

A. What we take from this

Three points carry past the particular model.

Graph structure is not informative merely because it is present, even when the generating process contains explicit spillover. The static intercept of Section III-B shows that spillover which is permanent and early is absorbed into the level of the receiving unit and becomes a fixed effect. A benchmark built that way will report that graph models do not help, and the report will be true of the benchmark and silent about the method. Anyone evaluating a spatio-temporal architecture should check that the spillover process is still moving inside the evaluation window.

The value of nonlinear message passing depends on the regime. In the untreated equilibrium a linear spatial lag cannot be distinguished from CG-CEN at ten seeds. Under intervention the gap opens to about 3.3 MAE with large effect sizes. Read practically, an agency running a steady-state monitoring model has little reason to prefer a graph network to classical spatial econometrics [5], [6], whereas one comparing prospective interventions with propagating consequences does.

Index design can settle a conclusion quietly. The growth-rate displacement index is internally coherent and, applied to a permanent shift, reports approximately nothing. An analyst who did not interrogate the functional form could reasonably conclude that inclusionary housing has no displacement consequence, when the index is simply unable to register a sustained level difference. The correction in (13) changes the measured contrast by two orders of magnitude without altering a single simulated quantity.

B. Limitations

No full-scale empirical evaluation. This is the principal limitation and we state it without hedging: every quantitative

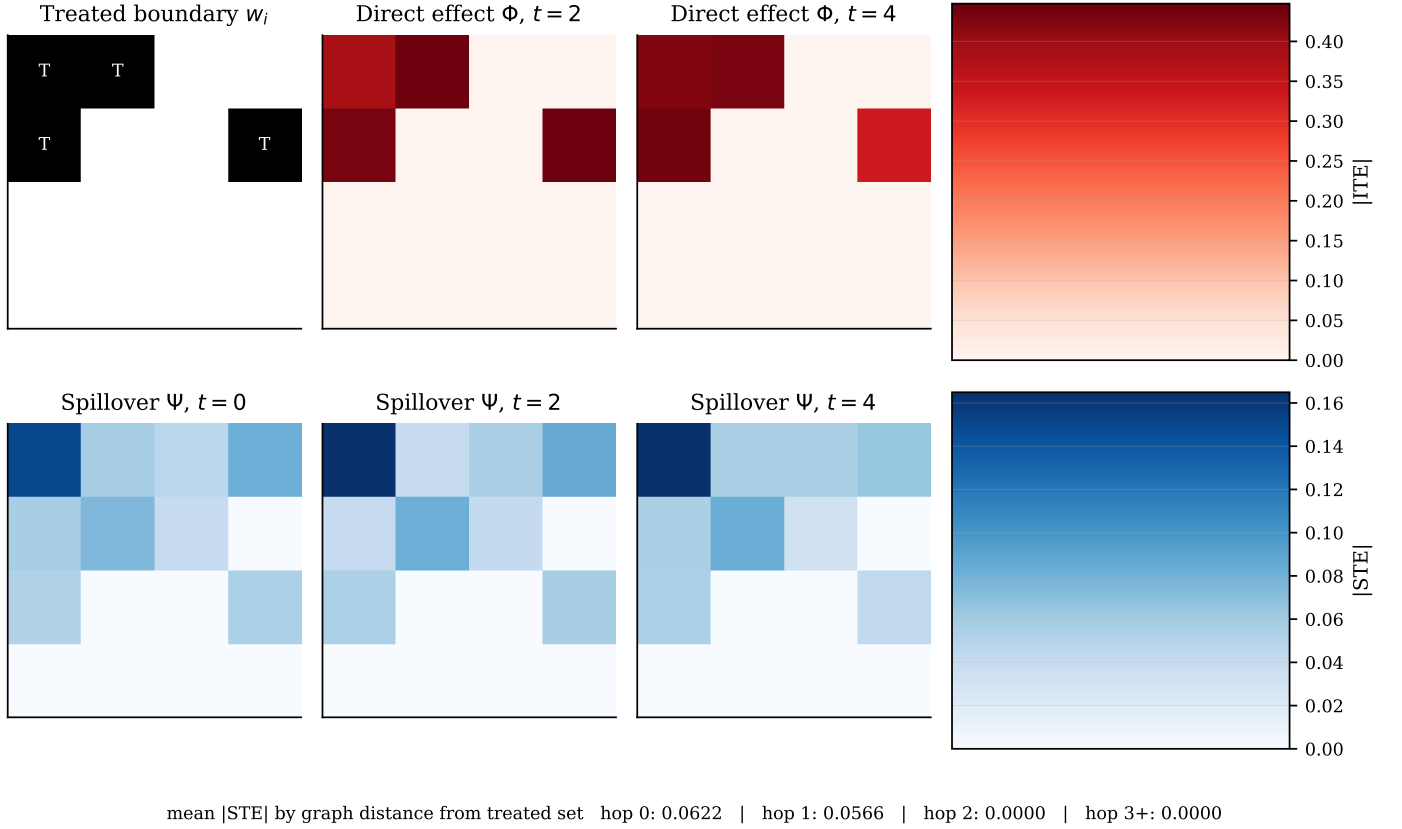


Fig. 2. Direct impact against multi-hop neighbour spillover on the 4×4 lattice, market-led scenario. Top row: the treated boundary and the magnitude of the treatment effect from Φ . Bottom row: the magnitude of the interference effect from the aggregated Ψ , at three evaluation steps. The treatment effect is exactly zero at untreated units and the interference effect is exactly zero at units with no treated neighbour, both by construction of (7) and (8).

result above comes from synthetic data. We implement a complete empirical pipeline, which ingests a census tract table carrying ACS and PLUTO fields, namely median gross rent B25064, median household income B19013, population B01003, housing units B25001, assessed land value, lot and residential area and a derived transit accessibility score, builds tract adjacency from centroid geodesics by k -nearest neighbours under a haversine metric, and emits a graph. We validate it end to end. But we validate it on a schema-compatible sample and not on a real extract. The loader reports its provenance through an explicit flag so that the distinction cannot be lost downstream, and no number derived from that sample appears anywhere in this paper.

The results therefore establish internal validity. Given a process with known dynamic spillover, the architecture recovers it, and we can say when it fails to. They establish nothing about external validity. Real tract panels are irregular, sparsely observed, subject to boundary revision between censuses, and carry confounding the generator does not simulate. Whether the conditional advantage documented here survives contact with such data is untested, and we would regard a policy application of CG-CEN before that test as premature.

Scale. Sixteen units over eighteen steps yields nine training windows. That is small, and it interacts with the results in a particular way: capacity is cheap to waste, which is part of why the temporal MLP is so variable, at ± 7.97 MAE, and why the linear model is competitive at all. Real metropolitan graphs

are larger, irregular, and have heavy-tailed degree distributions which a regular lattice does not exercise. We do not claim the ordering transfers unchanged.

Interventions. Policies are applied as multiplicative adjustments on fixed sets with fixed timing. Real interventions are heterogeneous in intensity, phased, politically endogenous to the neighbourhoods which receive them, and often anticipated by markets before enactment. Endogeneity in particular would break the exogeneity our design assumes, and addressing it needs identification machinery, whether instruments, differences in differences with spatial lags, or an explicit assignment model, which we have not built.

The exposure mapping. We assume interference travels only through the one-hop summary of (3). This excludes longer-range effects not mediated by intervening units, and it compresses a neighbourhood assignment vector into a scalar. Both are standard simplifications [2], [4] and both are falsifiable.

C. Open questions

The immediate priority is empirical replication. Obtaining a genuine tract extract and rerunning the benchmark against it is the single most valuable next step, and the loader exists so that this is a change of configuration rather than a piece of engineering. A useful intermediate would be semi-synthetic: real tract geometry, adjacency and initial conditions, with

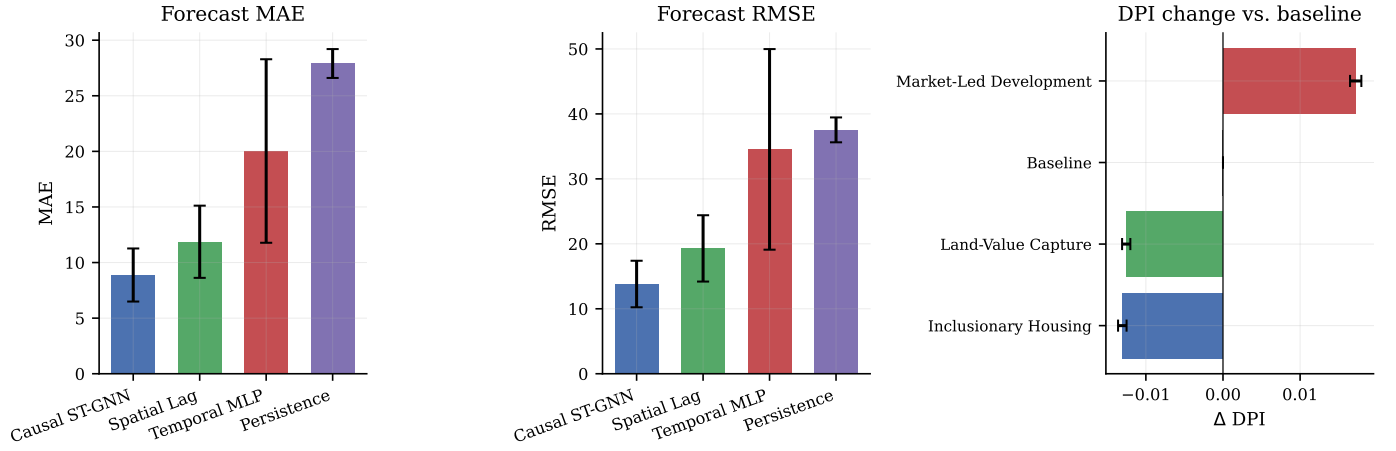


Fig. 3. Model families pooled over ten seeds and four scenarios, on the left and in the centre, and displacement pressure mitigation relative to the untreated baseline, on the right. Error bars give one standard deviation across seeds.

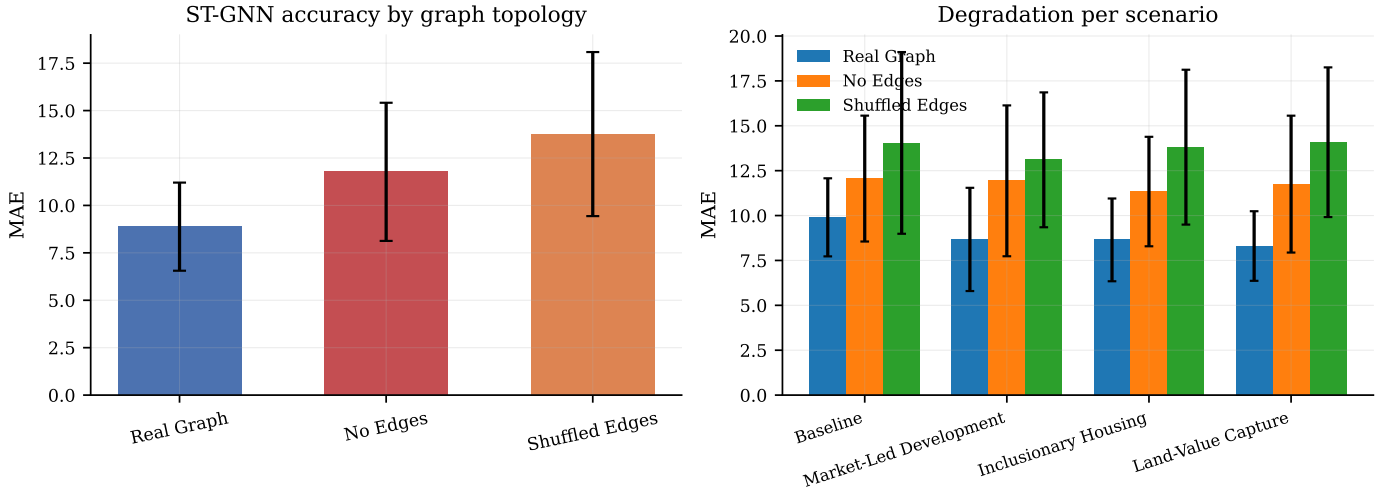


Fig. 4. Topology ablation. Left: pooled CG-CEN accuracy under the real graph, an edgeless graph, and a graph rewired by a random permutation of the units, which leaves the degree sequence unchanged. Right: the same broken down by scenario. Permuting the topology costs more than removing it, which indicates that the model has learned the particular adjacency rather than using message passing as a smoother.

simulated interventions layered on, which keeps the ground truth while introducing realistic structure.

Beyond that, three questions follow naturally. Richer exposure mappings would relax the one-hop scalar towards multi-hop or distance-weighted summaries, and it is not obvious how much this buys on a real adjacency. Uncertainty quantification over the estimated effects would let the scorecard report intervals on policy contrasts rather than point estimates; conformal and ensemble approaches are the obvious candidates. Modelling the assignment itself would carry the framework from the ex-ante setting, in which assignment is stipulated, towards observational evaluation, in which it has to be estimated.

D. Conclusion

We have presented CivicTwin, a framework which splits urban policy effects into a direct term and an interfering term under partial interference, with the split enforced by the architecture rather than by regularisation. On a controlled benchmark over ten seeds and four scenarios it outperforms

a spatial lag model, a graph-blind temporal MLP and persistence, and the topology ablations show that the graph is carrying the advantage, since permuting the adjacency is worse than removing it.

The advantage is conditional and we have tried to be exact about the condition. CG-CEN separates from classical spatial econometrics under intervention, at p between 0.0015 and 0.0033, and not in the untreated equilibrium, at $p = 0.127$. Under an earlier generator, in which the spillover arrived once and stayed, it separated from nothing at all, and neither did anything else, because spillover of that kind collapses into a static intercept. We report the negative result beside the positive one because for anyone building spatio-temporal policy models the boundary between the two regimes is more actionable than either result taken alone.

DATA AND CODE AVAILABILITY

The synthetic generator, the four-model baseline suite, the ten-seed benchmark harness, the topology ablations and the scripts which produce every figure and table in this paper

are available at <https://github.com/AbhayZ1/CivicTwin>. The repository also carries a configuration profile which reproduces the degenerate static-shock regime of Section III-B, so that the negative result reported there remains auditable.

REFERENCES

- [1] D. B. Rubin, “Randomization analysis of experimental data: The Fisher randomization test comment,” *J. Amer. Statist. Assoc.*, vol. 75, no. 371, pp. 591–593, 1980.
- [2] M. G. Hudgens and M. E. Halloran, “Toward causal inference with interference,” *J. Amer. Statist. Assoc.*, vol. 103, no. 482, pp. 832–842, 2008.
- [3] E. L. Ogburn and T. J. VanderWeele, “Causal diagrams for interference,” *Statist. Sci.*, vol. 29, no. 4, pp. 559–578, 2014.
- [4] L. Forastiere, E. M. Airoidi, and F. Mealli, “Identification and estimation of treatment and interference effects in observational studies on networks,” *J. Amer. Statist. Assoc.*, vol. 116, no. 534, pp. 901–918, 2021.
- [5] L. Anselin, *Spatial Econometrics: Methods and Models*. Dordrecht, The Netherlands: Kluwer Academic Publishers, 1988.
- [6] J. P. LeSage and R. K. Pace, *Introduction to Spatial Econometrics*. Boca Raton, FL, USA: Chapman and Hall/CRC, 2009.
- [7] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017.
- [8] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.
- [9] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in *Proc. 27th Int. Joint Conf. Artif. Intell. (IJCAI)*, 2018, pp. 3634–3640.
- [10] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, “Learning phrase representations using RNN encoder-decoder for statistical machine translation,” in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2014, pp. 1724–1734.
- [11] U. Shalit, F. D. Johansson, and D. Sontag, “Estimating individual treatment effect: Generalization bounds and algorithms,” in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 3076–3085.
- [12] S. Wager and S. Athey, “Estimation and inference of heterogeneous treatment effects using random forests,” *J. Amer. Statist. Assoc.*, vol. 113, no. 523, pp. 1228–1242, 2018.
- [13] Y. Ma and V. Tresp, “Causal inference under networked interference and intervention policy enhancement,” in *Proc. 24th Int. Conf. Artif. Intell. Statist. (AISTATS)*, 2021, pp. 3700–3708.
- [14] M. Zuk, A. H. Bierbaum, K. Chapple, K. Gorska, and A. Loukaitou-Sideris, “Gentrification, displacement, and the role of public investment,” *J. Planning Literature*, vol. 33, no. 1, pp. 31–44, 2018.
- [15] Center for Neighborhood Technology, “The housing and transportation (H+T) affordability index,” 2023. [Online]. Available: <https://htaindex.cnt.org/>